

Micro Project 1: Campus City Emergency Distribution

Course: 24MAT382 Computational Optimization And Applications

Github Repository: <https://github.com/Asher-Mathew/MICRO-PROJECT.git>

1 Problem Statement

Campus City Logistics currently operates an inefficient ad-hoc distribution system. This project designs a cost-optimized network to supply 15 campus facilities. The goal is to select exactly two warehouses that minimize total annual costs—including amortized construction, daily operations, and transportation—while ensuring all facility demands are met within a \$1.5 million budget.

2 Mathematical Formulation

The model uses Mixed-Integer Linear Programming (MILP) to find the global minimum cost.

2.1 Decision Variables

- $x_{w,f}$: Annual units shipped from warehouse w to facility f .
- y_w : Binary variable (1 if warehouse w is selected, 0 otherwise).

2.2 Objective Function

Minimize Total Cost (Z):

$$\text{Min } Z = \sum (T_{w,f} \cdot x_{w,f}) + \sum \left(\frac{\text{Const}_w}{10} + (\text{Op}_w \cdot 365) \right) y_w$$

2.3 Constraints

- **Demand:** All facilities must receive their required annual units.
- **Capacity:** Total shipments from a warehouse cannot exceed its annual limit.
- **Redundancy:** Exactly 2 warehouses must be active ($\sum y_w = 2$).
- **Financial:** Total cost Z must be $\leq \$1,500,000$.

3 Methodology

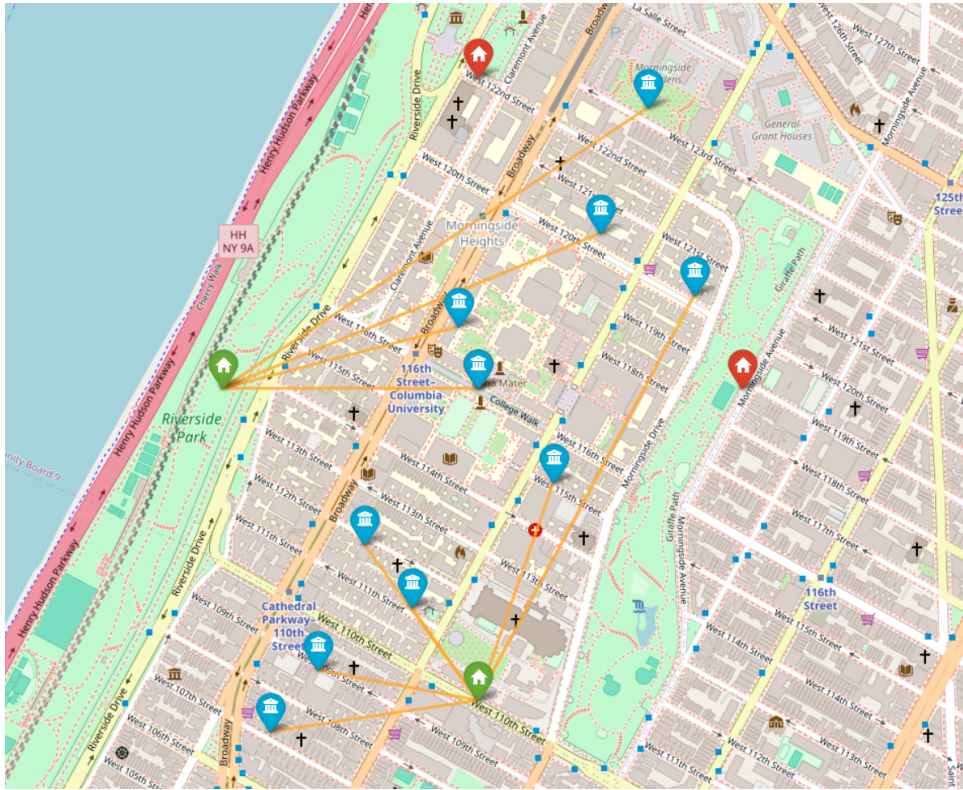
The analysis was performed using the **PuLP** library in Python to solve the MILP model. Geographic visualization was handled via **Folium** and **OSMnx**, which allowed us to route supplies along real-world campus streets rather than simple straight lines. This ensures the distances and costs used in the model reflect actual driving conditions.

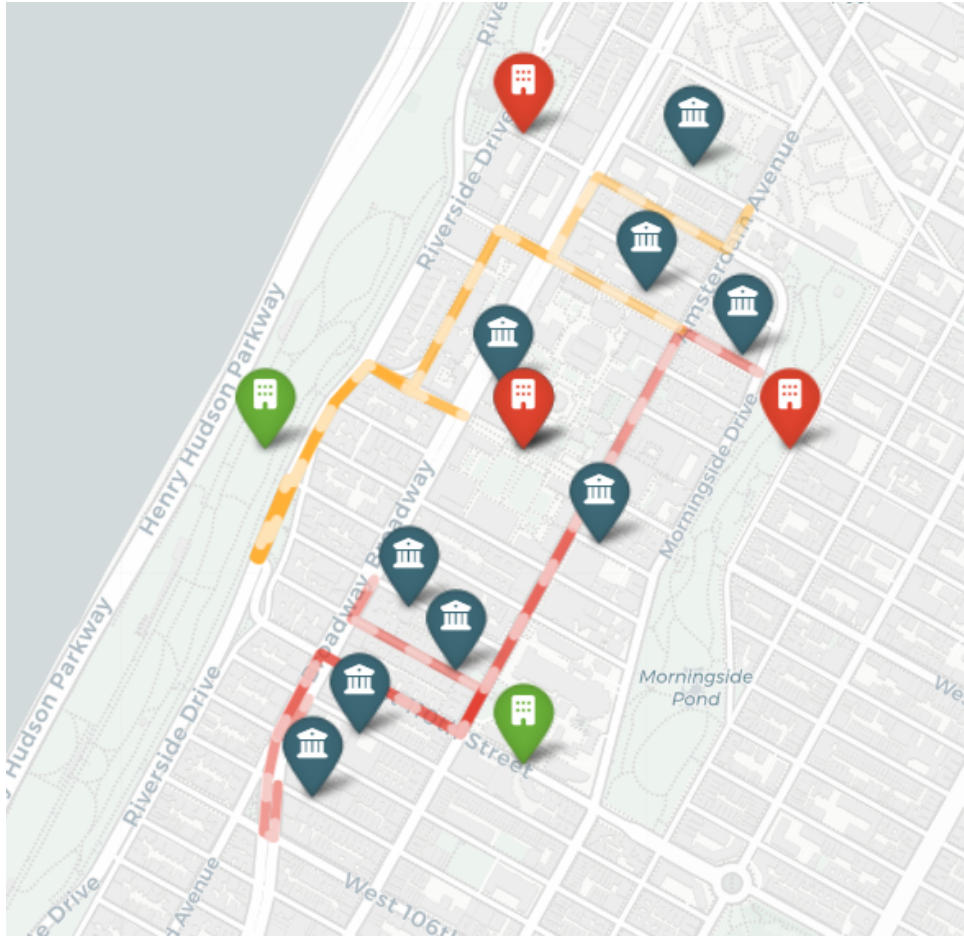
Source Warehouse	Target Facility	Annual Units
WH.NORTH	MED_CENTER	29,565
WH.NORTH	DORM_A	17,885
WH.SOUTH	DORM_B	18,615

Table 1: Optimal Shipment Plan Snippet

4 Results & Interpretation

The optimization successfully identified the most cost-effective warehouse pair. The selection was driven by the proximity to high-demand locations like the Medical Center and North Dormitory.





5 Conclusion

The proposed plan provides a 10-year sustainable distribution strategy. By utilizing road-network routing, we have verified that the \$1.5M budget is sufficient to maintain operations while meeting the critical needs of the campus.